

Hunar Intern - Data Science Internship

Assignment 1 Report

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Task 1: NumPy - Sum of Diagonal Elements

Objective: Create a square matrix using user input and calculate the sum of its diagonal elements using NumPy and a loop.

```
... Enter the order of the matrix (e.g., 2 for 2x2): 3
```

```
Enter the matrix elements:
```

```
Element [1][1] : 2
```

```
Element [1][2] : 4
```

```
Element [1][3] : 3
```

```
Element [2][1] : 3
```

```
Element [2][2] : 6
```

```
Element [2][3] : 4
```

```
Element [3][1] : 6
```

```
Element [3][2] : 8
```

```
Element [3][3] : 9
```

```
Matrix:
```

```
[[2 4 3]
```

```
 [3 6 4]
```

```
 [6 8 9]]
```

```
Sum of diagonal elements = 17
```

```

import numpy as np
|
n = int(input("Enter the order of the matrix (e.g., 2 for 2x2): "))

matrix = []
print("\nEnter the matrix elements:")

for i in range(n):
    row = []
    for j in range(n):
        value = int(input(f"Element [{i+1}][{j+1}] : "))
        row.append(value)
    matrix.append(row)

arr = np.array(matrix)

diagonal_sum = 0
for i in range(n):
    diagonal_sum += arr[i][i]

print("\nMatrix:")
print(arr)

print("\nSum of diagonal elements =", diagonal_sum)

```

Task 2: SQL

Created Student and Course tables, inserted records and executed the required SQL queries.

Student Table

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'hunar_intern' database and its tables. The main pane shows the SQL query history, which includes the following queries:

```

-- Query 23: Inserting data into the Student table
INSERT INTO Student (rollno, name, marks, aadhar_no, address)
VALUES
(101, 'Rohit', 65, '123456789104', 'Meerut'),
(105, 'Sohan', 20, '123456789105', 'Agra');

-- Query 24: Selecting all data from the Student table
SELECT * FROM Student;

-- Query 25: Inserting data into the Course table
INSERT INTO Course (rollno, course, course_duration)
VALUES
(101, 'BCA', '3 Years'),
(102, 'BBA', '3 Years'),
(103, 'BCA', '3 Years'),
(104, 'MCA', '2 Years'),
(105, 'BCA', '3 Years');

-- Query 26: Selecting all data from the Course table
SELECT * FROM Course;

-- Query 27: Calculating the average marks for each student
SELECT AVG(Marks) AS Average_Marks

```

The Data Output pane shows the results of the query 'SELECT * FROM Student;'. It displays 5 rows of data:

rollno	name	marks	aadhar_no	address
101	Rahul	85	123456789101	Delhi
102	Riya	92	123456789102	Noida
103	Aman	28	123456789103	Ghaziabad
104	Rohit	65	123456789104	Meerut
105	Sohan	20	123456789105	Agra

Course Table

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (3)
 - company
 - hunar_intern
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Login/Group Roles
 - Tablespaces

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Query Query History

```
26 SELECT * FROM Student;  
27  
28 INSERT INTO Course (RollNo, Course, Course_Duration)  
29 VALUES  
30 (101, 'BCA', '3 Years'),  
31 (102, 'BBA', '3 Years'),  
32 (103, 'BCA', '3 Years'),  
33 (104, 'MCA', '2 Years'),  
34 (105, 'BCA', '3 Years');  
35  
36 SELECT * FROM Course;  
37  
38 SELECT AVG(Marks) AS Average_Marks  
39 FROM Student;  
40  
41 SELECT Name
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	rollno	course	course_duration
	Integer	character varying (50)	character varying (20)
1	101	BCA	3 Years
2	102	BBA	3 Years
3	103	BCA	3 Years
4	104	MCA	2 Years
5	105	BCA	3 Years

Total rows: 5 Query complete 00:00:00.097 CRLF Ln 36, Col 1

Note: Query output screenshots (Average Marks, Sorting, Marks<30, Name starts with R, BCA students) can be appended if required.